biotray

BioTray vertical-farm tray controller: manufacturing manual

> **Status: design only; nothing built.** PCB: **order-ready: every automatic check passes**. Firmware control core host-tested (22 checks); ESP32 firmware compiles for the ESP32-S3 (arduino-cli, core 2.0.17, 0 warnings; `tools/esp32_build.sh`, binaries in `firmware/release/`), not yet run on hardware. pH/EC calibration values are placeholders.

1. What is made

A 100 x 80 mm controller: 12 V in, ESP32-S3 (WiFi/MQTT), four low-side 12 V outputs (2 pumps, fan, PWM grow light), DS18B20, I2C (BH1750/SHT31), water-level switch, analog pH and EC inputs.

2. Deliverables

Item	Where	State
PCB	`utility10/pcb/release/biotray/` and `biotray/hardware/`	**order-ready: every automatic check passes**
Control core + tests	`biotray/firmware/lib/bt_core.[ch]`, `test/test_bt.c`	passes
ESP32 skeleton	`biotray/firmware/src/main_biotray.cpp`	compiles for the ESP32-S3 (0 warnings), not run on hardware

3. Bill of materials

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
J2	12 V DC in	BarrelJack_CUI_PJ-063AH_Horizontal	CUI/Same Sky PJ-063AH	C22434582	hand/THT
F1	PTC 1A 16V	Fuse_1812_4532Metric	Littelfuse 1812L110/16DR	C142746	SMT (JLC)
D3	SMBJ15A	D_SMB	Littelfuse SMBJ15A	C83846	SMT (JLC)
Q5	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
D4	BZT52C12	D_SOD-123	DIODES BZT52C12-7-F	C124196	SMT (JLC)
R40	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
C30	100u/35V	CP_Elec_6.3x7.7	Lelon VZH101M1VTR-0607	C176683	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
C31	10u/50V	C_1210_3225Metric	Samsung CL32A106KBJNNNE	C380537	SMT (JLC)
U2	AP63205WU-7	TSOT-23-6	AP63205WU-7	C2071056	SMT (JLC)
R41	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
C32	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
L1	4.7u	L_Changjiang_FNR5040S	Changjiang FNR5040S4R7MT	C167963	SMT (JLC)
C33	22u/10V	C_0805_2012Metric	Samsung CL21A226MPQNNNE	C29277	SMT (JLC)
C34	22u/10V	C_0805_2012Metric	Samsung CL21A226MPQNNNE	C29277	SMT (JLC)
D5	SS14	D_SMA	SS14	C84343	SMT (JLC)
U3	AP2112K-3.3	SOT-23-5	AP2112K-3.3TRG1	C51118	SMT (JLC)
C35	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
C36	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
U1	ESP32-S3-WROOM-1-N8	ESP32-S3-WROOM-1	ESP32-S3-WROOM-1-N8	C2913198	SMT (JLC)
C37	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C38	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
R42	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
C39	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
SW2	Reset (EN)	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
SW3	Boot (IO0)	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
J1	USB-C	USB_C_Receptacle_HRO_TYPE-C-31-M-12	TYPE-C-31-M-12	C165948	SMT (JLC)
R43	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
R44	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
D6	Green	LED_0603_1608Metric	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	SMT (JLC)
R45	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
J7	Water temp DS18B20	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
R46	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
J8	I2C (BH1750 / SHT31)	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
R47	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
R48	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
J9	Water-level float switch	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
R49	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J10	pH module (0-3 V analog)	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
R50	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
C40	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
J11	EC module (0-3 V analog)	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
R51	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
C41	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
J12	UART flash/debug (USB-serial adapter)	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
Q1	AO3400A	SOT-23	AO3400A	C20917	SMT (JLC)
R10	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
R14	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
J3	Pump 1	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
D10	SS14	D_SMA	SS14	C84343	SMT (JLC)
Q2	AO3400A	SOT-23	AO3400A	C20917	SMT (JLC)
R11	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
R15	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
J4	Pump 2	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
D11	SS14	D_SMA	SS14	C84343	SMT (JLC)
Q3	AO3400A	SOT-23	AO3400A	C20917	SMT (JLC)
R12	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
R16	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
J5	Grow light (PWM)	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
Q4	AO3400A	SOT-23	AO3400A	C20917	SMT (JLC)
R13	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
R17	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
J6	Fan	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
D13	SS14	D_SMA	SS14	C84343	SMT (JLC)

Bought-in

12 V 3 A adapter (barrel 5.5/2.1, centre positive), 12 V peristaltic or submersible pumps, 12 V fan, 12 V LED grow strip (total under 1.1 A or replace the PTC), DS18B20 probe, BH1750 module, float switch, pH module and probe, EC module and probe, tubing, tray.

4. Make the board

1. Upload `utility10/pcb/release/biotray/1_fabrication/biotray_gerbers.zip` to the PCB fab (2 layers, 1.6 mm, ENIG or lead-free HASL, green mask, white silk; 1 oz is fine for loads up to about 1 A total). 2. For assembly upload `2_assembly/biotray_JLC_BOM.csv` and `biotray_JLC_CPL.csv`; check every polarity in the preview (diodes, electrolytic capacitor, USB-C, ESP32 module pin 1). 3. Through-hole connectors and pin sockets are listed with LCSC numbers but are normally hand-soldered. 4. General rules, tolerances and inspection: `manuals/STANDARD_PROCEDURES.md` sections A and B.

5. Program and configure

Arduino-ESP32 core 2.x with PubSubClient, OneWire, DallasTemperature, BH1750. Copy `secrets.h.example` to `secrets.h`. Calibrate pH with pH 4 and pH 7 buffers and EC with a standard solution, then edit `PH_CAL`, `EC_CAL`. MQTT topics: see `biotray/docs/DESIGN.md`.

6. Test (see `biotray/docs/TEST_PLAN.md`)

Supply rails, reverse-polarity test, dummy loads on every output, dry-run test (remove water: pump stops and latches), 48 h run.

7. Limits

Water, mains adapter and electricity together: use a GFCI-protected supply and an enclosure with proper sealing. No isolation on the pH probe. Not tested on hardware.